PROOF OF CONCEPT · DRAFT

Who Calls Themselves a *Learning Engineer?*

A scoping survey across LinkedIn, ResearchGate, Google Scholar, and institutional rosters to identify individuals who explicitly self-apply the title "Learning Engineer" — and characterize the emerging field they represent. **Not exhaustive**; results reflect reachable sources as of April 2026 and are intended to motivate a fuller study.

211	81	131	7	10	20+
TOTAL RECORDS	CONFIRMED LE TITLES	COMPANIES SEARCHED	DATA SOURCES	QUERIES RUN	COUNTRIES REPRESENTED

EXECUTIVE SUMMARY

"Learning Engineer" is an emerging, contested professional title rooted in Carnegie Mellon's Open Learning Initiative and the ICICLE consortium. This survey enumerates individuals who hold it explicitly — not instructional designers who happen to use the phrase. Of 211 records, **81 (38%)** are confirmed title-holders; another 102 (48%) are candidates pending profile verification.

> **Core finding:** The title is real, used globally, and growing — but means different things in different sectors. CMU METALS graduates form the most cohesive definitional community. Independent uses in EdTech, defense, healthcare, and corporate L&D vary significantly in scope. France is a surprising #2 by country (8%), driven by corporate L&D adoption of the English term.

WHY DISCOVERY IS HARD

Title fragmentation. Practitioners doing learning-engineering work use older labels (instructional designer, curriculum engineer, learning architect) — those won't surface in a title-keyword search.

ML contamination. "Machine / Deep / Robot / Reinforcement Learning Engineer" share the trigram. Auto-triage flags ~7–8% of LinkedIn hits as ML false-positives.

LinkedIn gatekeeping. API is not public; only PhantomBuster-style scrapes return structured data, and only headlines — full profiles require manual review.

Web snippet truncation. Brave API snippets are 150–200 chars and rarely reproduce LinkedIn headlines verbatim. Only 4 confirmed records came from 131 company searches.

SECTION 01 · METHODOLOGY

Search Strategy & Query Log

Full provenance in `titlesearch/PROVENANCE.md`. Raw outputs in `titlesearch/data/raw/`.

LI	LinkedIn via PhantomBuster — global "learning engineer" search across keyword and title-field variants with ML exclusions (machine/deep/robot/reinforcement); three query configurations pre-logged as Q004–Q006, ingested as a single merged seed	171 records
Q002	ResearchGate — site-scoped "learning engineer" with ML phrase exclusions	13 records
Q003	Google Scholar — "learning engineer" author/affiliation search (both pages)	4 records
Q007	Brave API — per-company web search (131 orgs, with ML exclusions)	1 record
Q008	Brave API — per-company web search (131 orgs, post-retrieval ML filter)	3 records
Q009	CMU METALS alumni page + ASU LEI team page (scrape)	7 + 2 upd
Q010	CMU OLI staff page + ASU faculty profile pages (56 requests)	2 + 2 upd
discard	*Earlier US-scoped LinkedIn pass (Q001) superseded by the global PhantomBuster runs above*	—

Caveat — proof-of-concept draft. This survey is a scoping exercise, not an exhaustive enumeration. The 171 LinkedIn records are reported as the aggregated PhantomBuster harvest (aggregation happened upstream of export). Coverage is known to be incomplete for non-English titles, government/defense practitioners, and older title variants (instructional designer, learning architect).

WHAT WORKED

METHOD	N	QUALITY
LinkedIn PhantomBuster (3 runs, dedup)	171	Medium
ResearchGate site search	16	High
Google Scholar affiliation	12	High
CMU METALS alumni directory	6+	High
Brave per-company web search	4	Medium

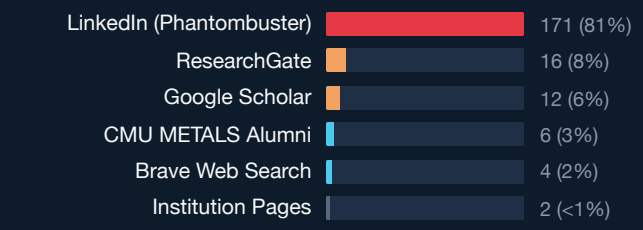
WHAT DIDN'T WORK

METHOD	REASON
ASU LEI team page	All 56 affiliates are faculty — research institute, not practitioner team
Company "Our Team" pages	~80% 404 or marketing-only; rarely list staff with titles
Brave -"phrase" exclusions	API doesn't reliably honor negative operators
LinkedIn without PhantomBuster	Scraping blocked; API requires partnership

SECTION 02 · FINDINGS

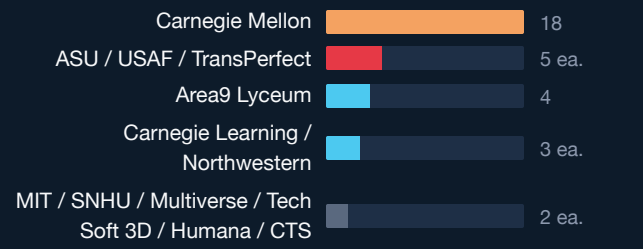
Who & Where: The LE Population

RECORDS BY SOURCE



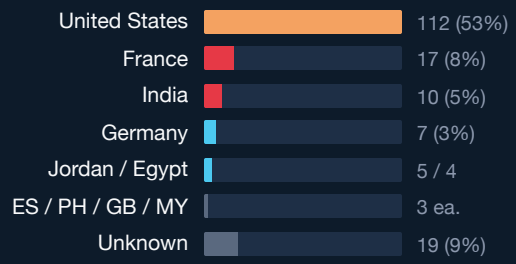
TRIAGE STATUS (N=211)

TOP ORGANIZATIONS



90+ orgs appear exactly once. CMU dominance reflects METALS pipeline + OLI.

GEOGRAPHY



TITLE VARIANTS

TITLE	N
Learning Engineer	27
Senior Learning Engineer	11
Digital / E-Learning Engineer	4
Developer Learning Engineer	2
Asst. Director of Learning Engineering	2
Chief / Lead / Staff / Associate LE (various)	5
Specialized (Medical, STEM, Robot, New Media)	5

COMPANIES SEARCHED · 131 TOTAL

EdTech 38 · Defense 22 · Consulting 16 · Higher Ed 12 · Healthcare 11 · Nonprofit 11 · Tech 7 · Finance 4 · HigherEd Infra 4 · Credentialing 4 · K-12 2

Career stages: practitioner 95 · industry 44 · unknown 61 · student 5 · faculty 4 · researcher 2.

SECTION 03 · PROPOSED TASK ANALYSIS

From Registry to KSA Framework

This registry is the foundation for a multi-method Cognitive Task Analysis (CTA) study to characterize what Learning Engineers actually *do* — the KSAs that define the role across sectors, and how they differ from instructional designers, ML engineers, and ed researchers.

PHASE 1

Stratified Sampling

Draw 15–20 confirmed practitioners from APPROVED, stratified by sector (EdTech, defense, higher ed, corporate) and career stage. Anchor on CMU METALS graduates; add independents for breadth.

PHASE 2A

Critical Incident Technique

Semi-structured 45–60 min interviews probing three episodes: success, challenge, ambiguous case. Extract task sequences, tools, decision triggers, tacit knowledge.

PHASE 2B

CTA Knowledge Probing

Knowledge Audit + Concept Mapping. Elicit participants' mental model of LE work; distinguish from adjacent titles; identify where they get stuck.

PHASE 3

Card Sort — KSA Taxonomy

Hybrid open+closed sort of 80–120 candidate KSA items from interview analysis. Reveals shared structure and cross-sector divergence.

PHASE 4

Survey Validation

Quantitative survey to full APPROVED + CANDIDATE pool (183 records). Validates KSA clusters; measures importance × frequency of task categories.

PHASE 5

KSA Synthesis

Integrate interview, card sort, and survey data into a formal KSA framework. Publish as reference for hiring rubrics, curriculum, certification (ICICLE / ISPI).

SECTION 04 · NEXT STEPS

IMMEDIATE DATA GAPS

102 CANDIDATE records unverified	manual LI review
61 records — career_stage unknown	title/org heuristics
19 records — no country code	org lookup
CMU METALS 2024–26 cohorts	re-scrape alumni
Google / MSFT / Amazon LE teams	conference rosters
ASU EdPlus staff (not LEI faculty)	separate scrape

RECOMMENDED NEXT QUERIES

- **Q011** ICICLE conference speaker/attendee lists (2019–2025)
- **Q012** LENS network member roster (JHU APL)
- **Q013** LinkedIn — title:"Learning Engineering" (mgmt layer)
- **Q014** Non-English variants: *ingénieur pédagogique*, *ingeniero de aprendizaje*
- **Q015** CMU METALS thesis DB — quantify LE placement rate

OPEN RESEARCH QUESTIONS

Q1 · Is there a coherent LE role?
Do CMU METALS grads, USAF LEs, and French "Digital Learning Engineers" share a meaningful KSA core — or are these three distinct professions using the same label?

Q2 · LE vs. Instructional Designer?
Many CANDIDATEs carry "Instructional Designer" as current title. Where do practitioners draw the line — and is the distinction perceived as meaningful?

Q3 · Is France a second epicenter?
17 French practitioners (8%) is disproportionate to France's share of global EdTech employment. Are there French-language training programs driving adoption of the English title?

Q4 · METALS pipeline rate?
If ~15–20% of METALS graduates take LE titles, it is the single most productive pipeline. How does this compare to MIT ODL, ASU LEI, Georgia Tech?